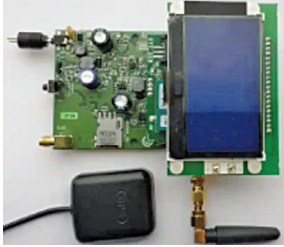
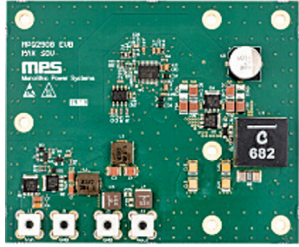


5 Interesting REFERENCE DESIGNS

Title of Reference Design	Reference Design For Smart Real-Time Vehicle Tracking System	Reference Design For 48V Buck Converter For Automotive Systems
Image		
Key Attraction	This reference design is for a smart vehicle tracking system based on GPS/GSM/GPRS. The reference design can predict accidents and car burglaries and share multiple data—such as location, speed, time, etc—over GPRS to the web server application.	This reference design is a high-voltage, synchronous, step-down controller that directly steps down voltages from up to 60V. The reference design can take an input from 4V to 60V and provide a constant output of 12V. It has a programmable soft start and multiple other safety features.
Highlights	The STEVAL-IEG001V2 from STMicroelectronics is a smart vehicle tracking system reference design. The reference design allows developers to easily create a GPS/GSM/GPRS based vehicle tracking system by providing a complete hardware and software solution with a clear and easy-to-use interface. It is powered by the STM32F401RE microcontroller, a powerful 32-bit Arm Cortex-M4 microcontroller. The reference design also includes a GSM/GPRS modem and GPS receiver, as well as a range of other interfaces and peripherals for connectivity and positioning. This unit features a 3-axis accelerometer and yaw/pitch/roll gyroscope L3GD20, which enables it to sense car lift using the MEMS. It can also predict vehicle theft by analysing the MEMS data. It can store the vehicle's status and location parameters in the microSD card and share the information to the admin by text message, thereby implementing black box functionality. The system consists of a vehicle unit (installed inside the vehicle) and remote web server application where the vehicle status and movement data can be viewed in an intuitive way on the map.	The MPQ2908 from Monolithic Power is a 48V to 12V buck converter capable of delivering 15A current and complying with CISPR-25 Class 5 EMC standards. The automotive buck converter has an efficiency of up to 95%. It employs a dual n-channel MOSFET driver. The reference design is AEC-Q100 qualified and has a wide 4V to 60V operating input range. It offers a programmable frequency range of 100kHz to 1000kHz, which can be programmed by an external resistor or synchronised to an external clock for noise-sensitive applications. The DC/DC converter has a programmable soft start and offers safety features, including output over-voltage protection (OVP), over-current protection (OCP), external input over-voltage protection (OVP), and external reverse-polarity protection (RPP). The SMPS also has a power good (PG) output voltage monitor and selectable cycle-by-cycle current limit that can improve the reliability of the connected devices.
Application	This reference design can be used for a wide range of vehicle tracking and fleet management applications. Moreover, it can predict crash and theft using the MEMS sensor.	This reference design is suitable for use in automotive power systems and industrial systems.
OEM Brand	STMicroelectronics	Monolithic Power Systems
URL	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-smart-real-time-vehicle-tracking-system	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-48v-buck-converter-for-automotive-systems